Alec Stanley

Aerospace Engineering (Hons) & Astrophysics · Monash University

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Fourth-year aerospace engineering and astrophysics student entering my final year in 2027, seeking **part-time engineering work** in design, simulation and testing alongside my studies. Full reports, code and media for each project are on my portfolio — codes like P-013 link to the project page.

Education

Bachelor of Engineering (Honours) / Bachelor of Science — Monash University 2023 – Expected Dec 2027
- Majors in **Aerospace Engineering** and **Astrophysics**; completed minors in Physics and Mathematics
- Distinction average (WAM 77.1); High Distinctions in CFD, spacecraft dynamics, astrophysics and programming

Ursula Frayne Catholic College — Perth WA · ATAR 98.65 2022
- Mathematics, physics, English and modern history; Year 12 Arts Representative; lead roles in musical theatre

Skills

Programming	Python (NumPy, Matplotlib, multiprocessing), MATLAB, C++, Mathematica, Arduino, Git
Simulation & Analysis	CFD (SU2), large-eddy simulation, aerodynamics, compressible flow, boundary layer theory, rigid-body and attitude dynamics, numerical integration, thermodynamics, photometry
Design & Fabrication	SolidWorks (CSWA, 235/240), 3D printing, laser cutting, electronics

Projects

Autonomous Rover — 38th Warman National Design and Build Competition P-009 May 2025
- Developed the electronics, control code and CAD in a team of 5 for a rover collecting 3 objects in 120 seconds
- Redesigned the drivetrain around mecanum wheels, a string-and-pulley telescoping arm and 2 ultrasonic sensors
- Built a custom Arduino control board and fabricated the chassis from laser-cut timber and 3D-printed parts

N-Body Gravitational Simulator HD — C++ P-017 Jun 2026
- Built a C++ gravitational simulator from scratch: a physics integration engine, plotting engine and window manager
- Implemented linked lists, vectors and arrays as custom data stores with manual memory management

Boundary Layers and Viscous Flow Regimes in CFD HD — SU2 P-014 P-015 Apr – May 2026
- Simulated laminar and turbulent flat-plate boundary layers at $\mathrm{Re} = 1.3 \times 10^6$ and 10×10^6, plus compressible flow over a 2D cylinder
- Validated boundary layer thickness and skin friction against Blasius theory, locating flow separation at 126.8°
- Ran a large-eddy simulation of a NACA 0012 airfoil to study flow regimes and its turbulent wake

Galaxy Collision Simulation HD — Python P-013 Oct 2025
- Integrated 242 gravitating bodies with a Leapfrog scheme to model the observed M51 and NGC 5195 collision, conserving total energy with angular momentum error of order 10^{-12}
- Cut runtime from tens of minutes to seconds using pre-allocated arrays, rendering 3 animations in parallel

Tumbling Rocket — Intermediate Axis Theorem HD — MATLAB · report 10/10 P-011 Oct 2025
- Propagated torque-free rotation of an arbitrary STL geometry, reproducing the Dzhanibekov effect
- Contrasted stable roll and yaw against chaotic 180° pitch reversals to justify spacecraft spin-axis choice

Glider Design, Build and Flight Performance — Design Build Fly team P-012 Oct 2025
- Developed a low-Reynolds glider over 3 iterations in a team of 5, using an Eppler E423 airfoil at aspect ratio 8
- Modelled lift, drag and stability in MATLAB, predicting a maximum glide ratio of 14.95, then compared to flight tests

Also on the portfolio: rocket launch and re-entry simulation (Python) P-010; shock structures in a supersonic wind tunnel P-007; distance and age of open cluster NGC 2175 P-016; exoplanet detection by transit photometry P-002

Experience

Resident Advisor — Richardson Hall, Monash University Feb 2024 – Jun 2025
- On-call first responder for a community of 200 residents, handling after-hours welfare and behavioural incidents with mental health first aid
- Coordinated community events and mediated conflict, escalating sensitive cases under duty-of-care procedures

Service Supervisor — Coles Supermarkets, Perth WA and Melbourne VIC Nov 2019 – Present
- Supervised front-end teams across 2 stores through peak trading, de-escalating customer and staff conflict
- Managed inventory, daily financial reporting and point-of-sale systems while training new team members